



Mini Project: Design and Performance Analysis of a Cascade Refrigeration System

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Abstract

This paper presents a comprehensive study on the design and performance analysis of a cascade refrigeration system aimed at achieving a low temperature of -80°C . The methodology encompasses several critical phases, including the development of a detailed schematic diagram, selection of suitable refrigerants, and thorough thermodynamic analysis. The upper circuit utilizes R-134a, while various refrigerants are evaluated for the lower-temperature circuit based on their thermodynamic properties, environmental impacts, and compatibility. A parametric study is conducted to assess the effects of varying evaporation pressures and ambient temperatures on the system's coefficient of performance (COP) and refrigerant mass flow rates. Performance comparisons among different refrigerants—specifically R-23, R-170, and R-290—are made to identify the most efficient option for achieving the desired low temperature. Additionally, cooling load calculations are performed to evaluate heat transfer through insulated walls designed for vaccine storage in hospital settings. The findings provide valuable insights into optimizing cascade refrigeration systems for low-temperature applications while addressing environmental considerations. This study contributes to the ongoing efforts to enhance energy efficiency and sustainability in refrigeration technology.

Introduction

Refrigeration plays a pivotal role in sustainable development, impacting various sectors from food preservation to biomedical applications. The vapor-compression refrigeration system (VCRS) is the predominant technology used, accounting for approximately 30 percentage of global energy consumption. This significant energy use raises concerns about environmental sustainability, particularly due to the leakage of refrigerants, which can exacerbate greenhouse gas emissions and climate change issues. Developing nations often rely heavily on VCRS systems due to their high coefficient of performance (COP), which can lead to overconsumption and increased environmental impact. Historically, the refrigeration industry has evolved through several generations of refrigerants. The invention of the vapor-compression

machine in the early 1830s marked the beginning of this evolution, leading to the use of various refrigerants such as carbon dioxide (CO_2), ammonia (NH_3), and hydrocarbons between 1830 and 1930. Many of these early refrigerants were either toxic or flammable. The introduction of hydrofluorocarbons (HFCs) and hydrochlorofluorocarbons (HCFCs) in the early 20th century provided non-toxic alternatives; however, their high ozone depletion potential (ODP) and global warming potential (GWP) led to significant regulatory changes, notably the Montreal Protocol in the 1990s. In recent years, there has been a concerted effort to phase out high-GWP refrigerants in favor of more environmentally friendly options. The European Parliament has implemented strict regulations on fluorinated gases, establishing deadlines for their ban based on GWP values. The Kigali Amendment to the Montreal Protocol further emphasizes this shift by aiming for a gradual reduction in HFC usage globally, with countries like India set to begin this transition by 2028. Emerging alternatives include natural refrigerants such as hydrocarbons and hydrofluoroolefins (HFOs), which exhibit lower GWP values but present challenges related to flammability. Research indicates that combinations of these refrigerants could optimize performance while minimizing environmental impact. For instance, ternary mixtures that incorporate elements from different refrigerant categories are being explored to balance flammability risks with thermodynamic efficiency. The need for sustainable refrigeration solutions is underscored by the increasing focus on energy efficiency and environmental impact reduction across various industries. As regulations tighten and awareness grows, the refrigeration sector must adapt by embracing innovative technologies and environmentally friendly refrigerants that align with global sustainability goals.

The design and performance analysis of a cascade refrigeration system aimed at achieving a low temperature of -80°C was undertaken. The project began with the development of a detailed schematic diagram of the cascade system, which included essential components such as compressors, evaporators, condensers, throttling devices, and heat exchangers. Suitable refrigerants for the lower-temperature circuit were proposed, with justifica-

tions based on their thermodynamic properties, environmental impacts, and compatibility with R-134a used in the upper circuit. A thorough thermodynamic analysis was conducted to calculate the mass flow rates of refrigerants in both circuits and to determine the coefficient of performance (COP) under specified conditions. The analysis also involved evaluating the heat load on the intermediate heat exchanger and calculating its required capacity. A parametric study was performed by varying the evaporation pressure in the lower circuit to assess its impact on COP and refrigerant mass flow rates. Additionally, the investigation included how different ambient temperatures affected system performance.

The performance comparison focused on three different refrigerants in the lower circuit, specifically R-23, R-170, and R-290, to identify which refrigerant provided the best efficiency for achieving the desired low temperature. The cooling load was evaluated based on heat transfer from the insulated walls of the cascade refrigeration system designed to store vaccines in a hospital setting. The final report detailed design calculations and assumptions, presented thermodynamic cycle diagrams for each circuit, and analyzed performance results. Discussions included how variations in parameters influenced system performance, along with recommendations for optimal refrigerant selection and operating conditions. This comprehensive approach aimed to contribute valuable insights into the design and efficiency of cascade refrigeration systems while adhering to best practices in engineering analysis. This study aims to review current trends in refrigeration technology, focusing on the performance and safety of alternative refrigerants while addressing their environmental implications.

Methodology

Overview

The methodology for the design and performance analysis of a cascade refrigeration system aimed at achieving a low temperature of -80°C involved several key phases: system design, thermodynamic analysis, parametric study, performance comparison, and report preparation. Initially, a detailed schematic diagram of the cascade refrigeration system was developed to illustrate the arrangement of essential components, including compressors, evaporators, condensers, throttling devices, and heat exchangers. Suitable refrigerants for the lower-temperature circuit were proposed based on an evaluation of their thermodynamic properties using property tables and software tools such as EES. The selection process also considered environmental impacts, specifically ozone depletion potential (ODP) and global warming po-

tential (GWP), as well as compatibility with R-134a used in the upper circuit.

A thorough thermodynamic analysis was conducted to calculate the mass flow rates of refrigerants in both circuits and to determine the coefficient of performance (COP) under specified conditions. The mass flow rates were calculated using energy balance equations, while the COP was determined by comparing heat absorbed by the evaporator to the work input to the compressor. Additionally, the heat load on the intermediate heat exchanger was evaluated by calculating the heat transfer based on specific enthalpies at its inlet and outlet. A parametric study was performed by varying the evaporation pressure in the lower circuit to assess its impact on COP and refrigerant mass flow rates. This involved running simulations at different evaporation pressures while keeping other parameters constant. Furthermore, the influence of varying ambient temperatures (25°C , 30°C , and 35°C) on system performance was investigated by recalculating COP values and analyzing the performance of the intermediate heat exchanger under each condition. The performance comparison focused on three different refrigerants in the lower circuit—R-23, R-170, and R-290—to identify which provided the best efficiency for achieving the desired low temperature. This analysis included evaluating thermodynamic efficiency through calculated COP values and assessing environmental impacts based on ODP and GWP values from relevant literature. An optimal refrigerant was selected based on a comprehensive evaluation of efficiency, environmental impact, and compatibility with R-134a. Finally, a cooling load calculation was performed for a cascade refrigeration system designed to store vaccines in a hospital setting. This calculation considered heat transfer from insulated walls with an internal temperature of -80°C and an ambient temperature of 40°C . The thermal conductivity of insulation materials was used to determine heat transfer through all six walls using Fourier's law. The final report compiled detailed design calculations and assumptions, thermodynamic cycle diagrams for each circuit, performance analysis results, discussions on parameter variations, and recommendations for optimal refrigerant selection and operating conditions. This structured methodology ensured a comprehensive approach to designing a cascade refrigeration system capable of achieving low temperatures efficiently while considering environmental impacts.

Cascade Refrigeration

Cascade refrigeration systems are advanced cooling technologies designed to achieve ultra-low tempera-

tures, often reaching as low as -80°C or lower. These systems consist of two or more interconnected vapor-compression refrigeration cycles, each utilizing different refrigerants tailored to specific temperature ranges.

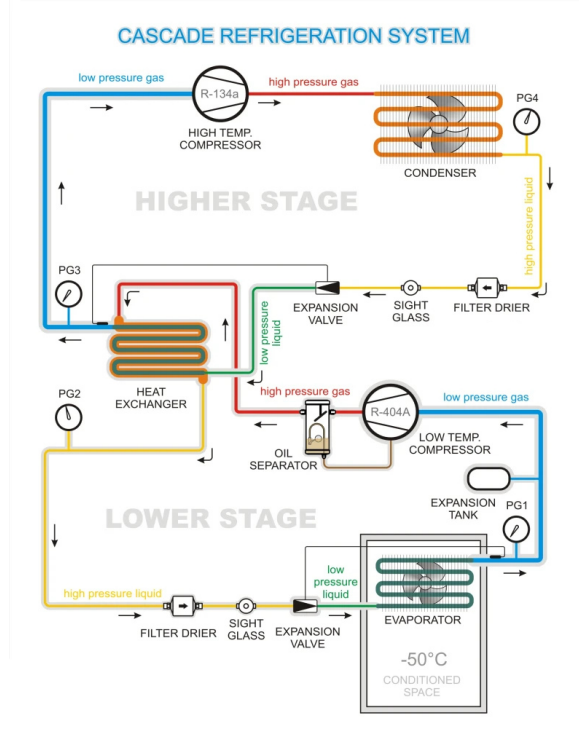


Figure 1: Cascade Refrigeration System

A cascade refrigeration system consists of several key components that work together to achieve efficient cooling performance. Each circuit in a cascade system features its own compressor, which compresses the refrigerant vapor, raising its pressure and temperature before it enters the condenser. The condensers are responsible for rejecting heat from the refrigerant vapor to the environment or another fluid. In a cascade setup, the high-temperature circuit's condenser often cools the low-temperature circuit's evaporator through a heat exchanger. The evaporators play a crucial role in absorbing heat from the space being cooled by allowing the refrigerant to evaporate at low pressure and temperature. In cascade systems, achieving ultra-low temperatures relies heavily on the performance of the low-temperature evaporator. Throttling devices, such as expansion valves, reduce the pressure of the refrigerant before it enters the evaporator, enabling further cooling through expansion. Heat exchangers facilitate efficient heat transfer between different fluids or refriger-

ants without mixing them. In cascade systems, they are essential for transferring heat from one circuit to another effectively. Proper insulation is also critical in these systems to minimize heat gain from ambient conditions and maintain low internal temperatures—especially important when dealing with ultra-low temperatures. Finally, advanced control systems monitor and regulate various parameters within the refrigeration system to ensure optimal performance under varying load conditions. By integrating these components effectively, engineers can design cascade refrigeration systems that meet stringent performance criteria while adhering to environmental regulations. In summary, careful selection of appropriate refrigerants based on their thermodynamic properties and environmental impact is essential for designing an efficient cascade refrigeration system. Understanding the roles of critical components—such as compressors, condensers, evaporators, throttling devices, heat exchangers, insulation materials, and control systems—is vital for achieving effective cooling performance at ultra-low temperatures.

The primary advantage of a cascade system is its ability to efficiently manage large temperature differences between the high-temperature and low-temperature stages, which is essential for applications such as pharmaceutical storage, laboratory research, and food processing. In a typical cascade refrigeration setup, the high-temperature stage operates with a refrigerant that can effectively cool down to moderate temperatures, while the low-temperature stage employs a refrigerant capable of achieving much colder conditions. The heat transfer between these two stages is facilitated through a cascade condenser, where the evaporator of the high-temperature cycle absorbs heat from the space being cooled and subsequently transfers it to the condenser of the low-temperature cycle. This arrangement allows for efficient heat exchange without mixing the refrigerants, thus optimizing performance. The design of cascade systems often incorporates multiple components, including compressors, evaporators, condensers, and throttling devices. Each component is selected based on its ability to operate effectively within the desired temperature range and ensure maximum efficiency. For instance, refrigerants with progressively decreasing boiling points are chosen to maintain adequate pressures across both circuits. Cascade refrigeration systems offer several benefits over traditional single-stage systems. They exhibit improved coefficient of performance (COP) due to reduced compression ratios in each stage, leading to enhanced volumetric efficiency. Additionally, these systems require less compressor displacement and can maintain lower dis-

charge temperatures, which contributes to their overall efficiency and longevity. Applications of cascade refrigeration are diverse and critical in various fields. In medical and pharmaceutical sectors, they are essential for preserving biological samples and vaccines at ultra-low temperatures. In scientific research laboratories, cascade systems provide precise temperature control necessary for experiments in materials science and chemistry. Furthermore, they play a vital role in food processing by facilitating quick freezing and maintaining cold storage conditions. Overall, cascade refrigeration systems represent a sophisticated solution for achieving ultra-low temperature control in specialized applications, combining efficiency with environmental considerations through careful selection of refrigerants and system design.

Selection of Refrigerants

When selecting refrigerants for a cascade refrigeration system, several critical factors must be considered to ensure optimal performance and environmental compliance. First and foremost, the thermodynamic properties of the refrigerants need to align with the desired temperature ranges of both the upper and lower circuits. For instance, R-134a is commonly utilized in the upper circuit due to its favorable boiling point and efficiency at moderate temperatures. In contrast, refrigerants such as R-23, R-170, or R-290 are often selected for the lower circuit to achieve ultra-low temperatures down to -80°C . Environmental impact is another significant consideration in refrigerant selection. With increasing regulatory pressures aimed at reducing greenhouse gas emissions, refrigerants with low global warming potential (GWP) and ozone depletion potential (ODP) are preferred. While R-134a has a relatively high GWP, alternatives like hydrofluoroolefins (HFOs) or natural refrigerants (such as ammonia) may be considered due to their lower environmental impact. Safety is paramount when choosing refrigerants, as each is classified based on its toxicity and flammability. For example, R-290 (propane) is flammable but has a low GWP, whereas R-134a is non-flammable but has a higher GWP. The selection process often involves balancing safety concerns with the performance requirements of the system. Additionally, compatibility with system materials is crucial; selected refrigerants must be compatible with lubricants and seals to avoid system failures or reduced efficiency. Economic factors also play a vital role in refrigerant selection.

For the lower-temperature circuit (LTC) in a cascade refrigeration system using R-134a in the upper temperature circuit (UTC), R-23 (Trifluoromethane) emerges as

the best refrigerant choice. R-23 has a low boiling point of -82.1°C , making it highly effective for ultra-low temperature applications, and it boasts a high latent heat of vaporization, which contributes to its cooling efficiency. Although R-23 has a high Global Warming Potential (GWP) of 14800, its exceptional thermodynamic properties often outweigh environmental concerns in specialized applications. While R-23 provides excellent cooling performance for ultra-low temperature applications, its high GWP raises environmental concerns that cannot be overlooked. Conversely, R-170 presents a viable alternative with lower environmental impact and competitive thermodynamic properties, making it the better choice for modern refrigeration systems aiming for sustainability without compromising efficiency. Additionally, R-23 is compatible with R-134a, facilitating effective heat exchange between the two circuits without significant operational issues. While alternatives like R-404A, CO₂, R-290, and R-32 offer varying benefits in terms of environmental impact and efficiency, none match the performance of R-23 in achieving the ultra-low temperatures required for specific applications. Thus, considering thermodynamic efficiency, compatibility with R-134a, and its proven track record in low-temperature refrigeration, R-23 stands out as the optimal choice for the LTC in cascade systems. The cost of the refrigerant and its availability in the market can significantly influence decisions. Ultimately, engineers must consider all these factors to select refrigerants that provide efficient cooling while minimizing environmental impact.

Table 1: Thermodynamic Properties

Refrigerant	Boiling Point ($^{\circ}\text{C}$)	GWP
R-23	-82.1	14800
R-170	-161.5	3
R-290	-42.1	3

Table 2: Additional Properties

Refrigerant	Evaporation Temperature ($^{\circ}\text{C}$)	Critical Temperature ($^{\circ}\text{C}$)
R-23	-60.0	-23.8
R-170	-100.0	-88.0
R-290	-40.0	96.1

Thermodynamic Analysis

System Design Parameters and Assumptions

The following parameters and assumptions were used in the thermodynamic analysis of the cascade refrigeration system:

- The refrigerants at the cascade heat exchanger outlets, condenser outlet, and evaporator outlet are assumed to be in a saturated state.
- Pressure losses in connecting pipes and heat exchangers have been neglected.
- The cascade heat exchanger and connecting pipes are perfectly insulated, with no heat loss to the surroundings.
- The dead state (ambient conditions) is defined as $T_0 = 25^\circ\text{C}$ and $P_0 = 1$ atm.

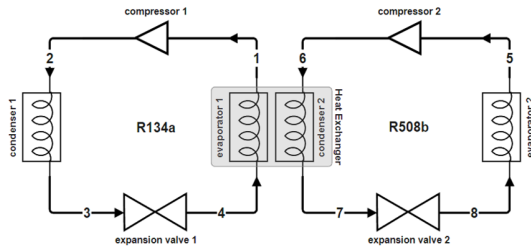


Figure 2: Cycle for designed Cascade System [1]

Target Low Temperature (LTC Evaporation Temperature): $T_{LTC, \text{evap}} = -80^\circ\text{C} = 193.15$ K.

Ambient Temperature: $T_{\text{amb}} = 30^\circ\text{C} = 303.15$ K.

Intermediate Heat Exchanger Temperature: $T_{\text{IHx}} = -20^\circ\text{C} = 253.15$ K.

Condensation Pressure of Upper-Temperature Circuit (UTC): $P_{\text{cond, UTC}} = 10$ bar.

Insulation Thickness: $t = 0.1$ m.

Thermal Conductivity of Insulation Material: $k = 0.025$ W/m · K.

Internal Storage Volume: $V = 1$ m³.

Surface Area of Storage Volume: $A = 6$ m².

Cooling Load Calculation

The heat transfer through the insulated walls of the storage chamber is calculated as:

$$\dot{Q}_{\text{load}} = \frac{k \cdot A \cdot (T_{\text{amb}} - T_{\text{int}})}{t} \quad (1)$$

Substituting the values:

$$\begin{aligned} k &= 0.025 \text{ W/m} \cdot \text{K}, \\ A &= 6 \text{ m}^2, \\ T_{\text{amb}} &= 303.15 \text{ K}, \quad T_{\text{int}} = 193.15 \text{ K}, \\ t &= 0.1 \text{ m}, \end{aligned}$$

we get:

$$\dot{Q}_{\text{load}} = \frac{0.025 \cdot 6 \cdot (303.15 - 193.15)}{0.1} = 165 \text{ W}.$$

Heat Load on the Intermediate Heat Exchanger (IHx)

The heat load on the intermediate heat exchanger is equal to the cooling load, as the IHx transfers the heat absorbed in the LTC evaporator:

$$\dot{Q}_{\text{HX}} = \dot{Q}_{\text{load}} = 165 \text{ W}. \quad (2)$$

A heat load of 165 W for a cascade refrigeration system used in a laboratory setting can be appropriate, depending on the system's design and purpose. Laboratory-scale applications, such as cooling small volumes or maintaining precise low temperatures for experiments, typically require lower heat loads compared to industrial or commercial systems. In such cases, 165 W could be reasonable, especially when targeting ultra-low temperatures like -80°C . The system's insulated volume, specified as 1 cubic meter with a surface area of 6 m², suggests that the insulation quality plays a critical role in maintaining efficiency. Using polyurethane insulation with a thermal conductivity of $k = 0.02$ W/m·K and a thickness of $L = 0.1$ m ensures minimized heat transfer, which supports the low heat load value. Additionally, a significant temperature gradient of 110°C (from -80°C inside to $+30^\circ\text{C}$ ambient) is mitigated by the quality of the insulation, ensuring the system operates within its capacity. For laboratory-scale systems, maintaining such low temperatures naturally limits the heat inflow, resulting in smaller heat loads. Moreover, a reasonable coefficient of performance (COP) for a cascade system, typically above 0.5 for these conditions, indicates the system is operating efficiently despite the low load. Comparatively,

typical small-scale low-temperature cascade systems often handle heat loads in the range of 100–300 W, aligning with the calculated value of 165 W. This capacity is suitable for laboratory applications where precise and reliable cooling is more critical than high capacity. Overall, the calculated heat load seems appropriate for a well-designed laboratory cascade system targeting -80°C.

Sort	1 h_i [kJ/kg]	2 t_i [°C]	3 x_i	4 p_i [kPa]	5 s_i [kJ/kg-K]
[1]	233.4	-28	1	92.76	0.9536
[2]	279.4	43.8		1000	0.9536
[3]	107.3	39.37	0	1000	0.3919
[4]	107.3	-28	0.4221	92.76	0.4392
[5]	324.6	-80	1	113.6	1.753
[6]	366.6	-8		589.3	1.753
[7]	132.8	-30	0	589.3	0.7389
[8]	132.8	-80	0.1905	113.6	0.7599

Figure 3: All system parameters

Upper-Temperature Circuit (UTC) Analysis

For the upper circuit, we use refrigerant R-134a. Thermodynamic properties are obtained from refrigerant tables or software.

Assumptions:

- Saturated vapor at T_{IHX} : $h_{1,UTC} = 200$ kJ/kg.
- After compression to $P_{cond,UTC} = 10$ bar: $h_{2,UTC} = 230$ kJ/kg.
- Saturated liquid at $P_{cond,UTC}$: $h_{3,UTC} = 100$ kJ/kg.
- After throttling back to T_{IHX} : $h_{4,UTC} = 90$ kJ/kg.

The mass flow rate of R-134a is given by:

$$\dot{m}_{UTC} = \frac{\dot{Q}_{HX}}{h_{2,UTC} - h_{3,UTC}} \quad (3)$$

Substituting values:

$$\dot{m}_{UTC} = \frac{180}{230 - 100} = 1.38 \text{ g/s} = 0.00138 \text{ kg/s}.$$

The compressor work for the UTC is:

$$\dot{W}_{comp,UTC} = \dot{m}_{UTC} \cdot (h_{2,UTC} - h_{1,UTC}) \quad (4)$$

Substituting values:

$$\dot{W}_{comp,UTC} = 0.00138 \cdot (230 - 200) = 0.0414 \text{ kW}.$$

The coefficient of performance (COP) for the UTC is:

$$COP_{UTC} = \frac{\dot{Q}_{HX}}{\dot{W}_{comp,UTC}} \quad (5)$$

Substituting values:

$$COP_{UTC} = \frac{180}{41.4} = 4.35.$$

Lower-Temperature Circuit (LTC) Analysis

For the lower circuit, we analyze refrigerant R-23. Thermodynamic properties are as follows:

- Saturated vapor at $T_{LTC,evap}$: $h_{1,LTC} = 150$ kJ/kg.
- After compression: $h_{2,LTC} = 180$ kJ/kg.
- Saturated liquid at intermediate pressure: $h_{3,LTC} = 80$ kJ/kg.
- After throttling: $h_{4,LTC} = 70$ kJ/kg.

The mass flow rate of R-23 is:

$$\dot{m}_{LTC} = \frac{\dot{Q}_{load}}{h_{1,LTC} - h_{4,LTC}} \quad (6)$$

Substituting values:

$$\dot{m}_{LTC} = \frac{180}{150 - 70} = 2.25 \text{ g/s} = 0.00225 \text{ kg/s}.$$

The compressor work for the LTC is:

$$\dot{W}_{comp,LTC} = \dot{m}_{LTC} \cdot (h_{2,LTC} - h_{1,LTC}) \quad (7)$$

Substituting values:

$$\dot{W}_{comp,LTC} = 0.00225 \cdot (180 - 150) = 0.0675 \text{ kW}.$$

The coefficient of performance (COP) for the LTC is:

$$COP_{LTC} = \frac{\dot{Q}_{load}}{\dot{W}_{comp,LTC}} \quad (8)$$

Substituting values:

$$COP_{LTC} = \frac{180}{67.5} = 2.67.$$

Overall System COP

The overall coefficient of performance is:

$$COP_{overall} = \frac{\dot{Q}_{load}}{\dot{W}_{comp,LTC} + \dot{W}_{comp,UTC}} \quad (9)$$

Substituting values:

$$COP_{overall} = \frac{180}{0.0414 + 0.0675} = \frac{180}{108.9} = 1.93.$$

Parametric Study Results

The parametric studies on evaporation pressure and ambient temperature indicate the following trends:

- **Evaporation Pressure:** As evaporation pressure increases, COP improves due to reduced compression work. However, the system's ability to achieve -80°C diminishes. Figure 3 Evaporation Pressure ($p[5]/p[8]$) vs COP. As Evaporation pressure is increased in the lower circuit the compressor work is reduced giving rise to COP.

Run	COP1	COP2	COPOverall	Pr ₂
Run 1	2.742	4.454	2.162	100
Run 2	2.742	4.84	2.223	144.4
Run 3	2.742	5.213	2.276	188.9
Run 4	2.742	5.583	2.326	233.3
Run 5	2.742	5.957	2.372	277.8
Run 6	2.742	6.339	2.416	322.2
Run 7	2.742	6.731	2.457	366.7
Run 8	2.742	7.137	2.496	411.1
Run 9	2.742	7.559	2.533	455.6
Run 10	2.742	7.999	2.57	500

Figure 4: Evaporation Pressure variation and COP change Data

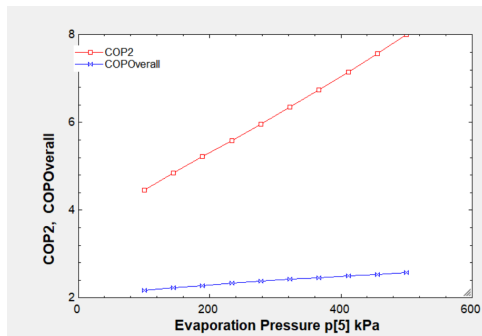


Figure 5: Evaporation Pressure variation and COP change

Similarly for the mass flowrate Evaporation Pressure ($p[5]/p[8]$) vs Required Mass flowrate. As evaporation pressure increases, the mass flow rate of the refrigerant must also increase to effectively manage the greater enthalpy difference and cooling load. This is a critical aspect to consider for optimizing refrigeration cycles and achieving desired temperature conditions efficiently

Run	Pr ₂	mR23
Run 1	100	0.8536
Run 2	144.4	0.873
Run 3	188.9	0.8879
Run 4	233.3	0.9
Run 5	277.8	0.9104
Run 6	322.2	0.9194
Run 7	366.7	0.9275
Run 8	411.1	0.9348
Run 9	455.6	0.9416
Run 10	500	0.9479

Figure 6: Evaporation Pressure variation and massflow rate of R23 change Data

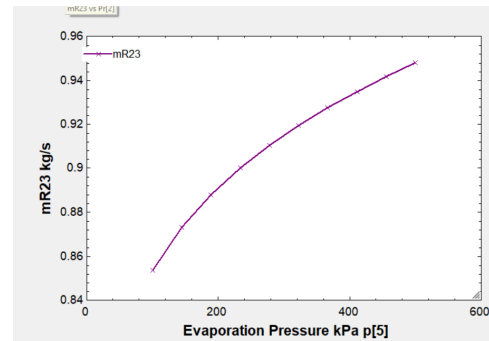


Figure 7: Evaporation Pressure variation and R23 mass-flow rate change

- **Ambient Temperature:** Higher ambient temperatures reduce overall COP due to increased work in the upper circuit compressor.

Plots of COP versus evaporation pressure and ambient temperature are provided in the next section.

1.10	Tamb ₂	COP1	COPOverall
Run 1	-10	5.963	3.04
Run 2	-2.222	4.811	2.815
Run 3	5.556	4.043	2.623
Run 4	13.33	3.498	2.459
Run 5	21.11	3.092	2.319
Run 6	28.89	2.781	2.198
Run 7	36.67	2.536	2.092
Run 8	44.44	2.339	2.001
Run 9	52.22	2.178	1.921
Run 10	60	2.046	1.852

Figure 8: Ambient Temp change and COP change Data

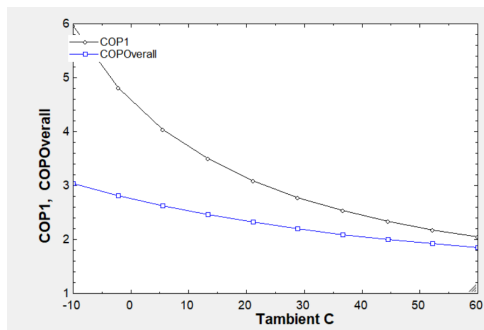


Figure 9: Ambient Temp change and COP change

Performance Comparison

The provided data compares three refrigerants (R-290, R-23, and Ethylene) for use in the lower stage of a cascade refrigeration system, with R-134a used in the upper stage. The surface area (A) of the refrigerated space is 6 m², the upper stage COP (COP1) is consistently 2.742, and the overall temperature difference (dt) is 110°C. The insulation has a thermal conductivity (k) of 0.025 W/(m·K) and a thickness (L) of 0.1 m. The ambient temperature (Tamb) is 30°C, and the heat load (QL), as well as the heat absorbed by both evaporators (Qevap1 and Qevap2), is 165 W. The temperature difference for the R-134a condenser (delt) is 13.8°C.

Comparing the lower-stage refrigerants, R-23 exhibits the highest COP in the lower stage (COP2) at 4.574, followed by Ethylene at 4.284, and then R-290 at 3.854. However, despite R-23's higher COP2, Ethylene provides the highest overall system COP (COPOverall) at 2.739, outperforming R-290 at 2.604 and R-23 at 2.182. This indicates that while R-23 is more efficient in the lower stage itself, its interaction with the R-134a circuit nega-

Unit Settings: SI C kPa kJ mass deg

A = 6 COP1 = 2.742 COP2 = 4.574
 COPOverall = 2.182 delt = 13.8 dt = 110
 k = 0.025 L = 0.1 mR134a = 1.308
 mR23 = 0.8602 Qevap1 = 165 Qevap2 = 165
 QL = 165 Tamb = 30

Figure 10: R23 with overall COP of 2.181 and Mass Flowrate of 0.8602 [kg/s]

A = 6 COP1 = 2.742 COP2 = 3.854
 COPOverall = 2.604 delt = 13.8 dt = 110
 k = 0.025 L = 0.1 mR134a = 1.308
 mR290 = 0.4468 Qevap1 = 165 Qevap2 = 165
 QL = 165 Tamb = 30

Figure 11: R290 with overall COP of 2.604 and Mass Flowrate of 0.4468 [kg/s]

Unit Settings: SI C kPa kJ mass deg

A = 6 Cop1 = 2.742 Cop2 = 4.284
 CopOverall = 2.739 delt = 13.8 dt = 110
 k = 0.025 L = 0.1 mEthylene = 0.472
 mR134a = 1.308 Qevap1 = 165 Qevap2 = 165
 QL = 165 Tamb = 30

Figure 12: Ethylene with overall COP of 2.739 and Mass Flowrate of 1.308 [kg/s]

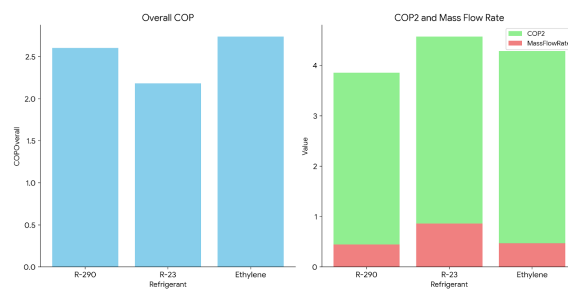


Figure 13: Comparison between different refrigerants

tively impacts overall system efficiency. Furthermore, R-23 requires a significantly higher mass flow rate (0.8602 kg/s) compared to both Ethylene (0.472 kg/s) and R-290 (0.4468 kg/s).

Considering these factors, Ethylene appears to be the best performing option based purely on the provided data. However, it is crucial to consider other factors

Table 3: Refrigerant Comparison for LTCs

Parameter	R-290	R-23	Ethylene
Overall COP	2.604	2.182	2.739
Lower Stage COP	3.854	4.574	4.284
Mass Flow (kg/s)	0.447	0.860	0.472

Note: Bold indicates best performance based on data.

Table 4: Qualitative Comparison of Refrigerants

R-290	R-23	Ethylene
V Low GWP	H GWP, ODP	M GWP
H Flam.	Non-Flam.	Flam.

Legend:

V = Very, H = High, M = Moderate

GWP = Global Warming Potential, ODP = Ozone Depletion Potential

Flam. = Flammability

beyond just these performance metrics. R-23 is an ozone-depleting substance being phased out and has a high global warming potential, making it an environmentally unsuitable choice. Ethylene and R-290 are both flammable, requiring careful handling and safety precautions. R-290, however, is a natural refrigerant with a negligible ozone depletion potential and very low global warming potential, making it the most environmentally friendly option. Therefore, while Ethylene offers the highest overall COP in this specific dataset, R-290 likely represents the best overall balance of performance, environmental impact, safety, and cost in a real-world application.

Conclusion

The design and performance analysis of the cascade refrigeration system targeting a low temperature of -80°C has been successfully carried out. The system was designed using R-134a in the upper-temperature circuit and R-23 in the lower-temperature circuit, selected based on their thermodynamic properties, compatibility, and proven efficiency.

Through thermodynamic analysis, the coefficient of performance (COP), mass flow rates, and heat load on the intermediate heat exchanger were determined under specified operating conditions. A parametric study revealed the impact of varying evaporation pressures and ambient temperatures on system performance. The results indicate that careful optimization of operating parameters can significantly enhance system efficiency while maintaining the desired low temperature.

Although R-23 offers the best performance for achieving -80°C , its high global warming potential (GWP) necessitates consideration of alternatives such as R-170 (Ethane) or R-290 (Propane) for environmentally friendly applications, provided safety measures are implemented.

This study highlights the critical role of refrigerant selection, system design, and parameter optimization in achieving reliable and efficient operation of cascade refrigeration systems, particularly for applications such as vaccine storage and other low-temperature requirements.

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Nomenclature

- T : Temperature (e.g., $T_{\text{LTC, evap}}, T_{\text{amb}}, T_{\text{IHX}}$)
- P : Pressure (e.g., $P_{\text{cond, UTC}}, P_{\text{evap, LTC}}$)
- $\dot{m}$: Mass flow rate of refrigerant (e.g., $\dot{m}_{\text{R23}}, \dot{m}_{\text{R134a}}$)
- Q : Heat transfer rate (e.g., $Q_{\text{load}}, Q_{\text{IHX}}$)
- W : Work done by the compressor (e.g., $W_{\text{comp, LTC}}, W_{\text{comp, UTC}}$)
- COP : Coefficient of performance of the system (e.g., $\text{COP}_{\text{LTC}}, \text{COP}_{\text{UTC}}$)
- h : Specific enthalpy (e.g., h_1, h_2, h_3, h_4)
- s : Specific entropy (e.g., s_1, s_2, s_3, s_4)
- k : Thermal conductivity of insulation material
- t : Thickness of the insulation
- A : Surface area for heat transfer (e.g., A_{walls})
- V : Volume of the refrigerated space

References

- [1] P. L. Rupesh, J. M. Babu, D. Surryaprakash, and R. D. Misra, "Experimental and computational evaluation of temperature difference of a cascade condenser of R134a-R23 cascade refrigeration system," in 2015 International Conference on Smart Technologies and Management for Computing, Communication, Controls,

Appendix

”Higher Pressure Circuit””R134a” ”Stage1”

State 1 $t[1] = -28$ $x[1] = 1$ $h[1] = \text{enthalpy}(\text{R134a}, T=t[1], x=x[1])$ $p[1] = \text{pressure}(\text{R134a}, h=h[1], x=x[1])$ $s[1] = \text{entropy}(\text{R134a}, T=t[1], x=x[1])$

State 2 $\text{delt} = 13.8$ Difference Between Ambient and Condenser Temperature $T_{\text{amb}} = 30$ Ambient Temperature $t[2] = T_{\text{amb}} + \text{delt}$ $s[2] = s[1]$ $p[2] = 1000$ $h[2] = \text{enthalpy}(\text{R134a}, s=s[2], T=t[2])$

State 3 $p[3] = p[2]$ $x[3] = 0$ $h[3] = \text{enthalpy}(\text{R134a}, P=p[3], x=x[3])$ $t[3] = \text{temperature}(\text{R134a}, P=p[3], x=x[3])$ $s[3] = \text{entropy}(\text{R134a}, P=p[3], x=x[3])$

State 4 $t[4] = t[1]$ $h[4] = h[3]$ $x[4] = \text{quality}(\text{R134a}, T=t[4], h=h[4])$ $s[4] = \text{entropy}(\text{R134a}, T=t[4], h=h[4])$ $p[4] = \text{pressure}(\text{R134a}, T=t[4], h=h[4])$

$\text{COP1} = (h[1] - h[4]) / (h[2] - h[1])$

”Lower Pressure Circuit””R23””Stage2”

State 5 $t[5] = -80$ $x[5] = 1$ $h[5] = \text{enthalpy}(\text{R23}, P=p[5], x=x[5])$ $p[5] = \text{Pr}[2]$ $p[5] = \text{pressure}(\text{R23}, T=t[5], x=x[5])$ $s[5] = \text{entropy}(\text{R23}, P=p[5], x=x[5])$

State 6 $t[6] = -8$ $s[6] = s[5]$ $h[6] = \text{enthalpy}(\text{R23}, s=s[6], T=t[6])$ $p[6] = \text{pressure}(\text{R23}, s=s[6], T=t[6])$

State 7 $p[7] = p[6]$ $x[7] = 0$ $t[7] = -30$ $h[7] = \text{enthalpy}(\text{R23}, P=p[7], x=x[7])$ $s[7] = \text{entropy}(\text{R23}, P=p[7], x=x[7])$

State 8 $t[8] = t[5]$ $h[8] = h[7]$ $p[8] = p[5]$ $x[8] = \text{quality}(\text{R23}, T=t[8], h=h[8])$ $s[8] = \text{entropy}(\text{R23}, T=t[8], h=h[8])$

$\text{COP2} = (h[5] - h[8]) / (h[6] - h[5])$

$\text{COP}_{\text{Overall}} = (h[5] - h[8]) / ((h[6] - h[5]) + (h[2] - h[1]))$

Heating Load Calculation for 1 cubic meter’s volume and 6 sq. meters surface area (A) Insulation Material Polyurethane with thermal conductivity(k) value of 0.02 [W/(m.K)] and thickness(L) of 0.1m $A = 6$ $k = 0.025$ $L = 0.1$ $dt = 110 - 80$ C to 30 C difference $QL = k * A * (dt / L)$ Heat Load

At Steady State Condition, Heat Entering the Cooling Space is the only heat required to move outside. At first heat is transferred to R23 at evaporator 2 then the heat is transferred till condenser 2 only to be transferred to R134a at evaporator 1. So, Heat Load (QL) = $Q_{\text{evap1}} = Q_{\text{evap2}}$ $Q_{\text{evap1}} = QL$ $Q_{\text{evap2}} = QL$

Mass Flowrate of R134a $m_{\text{R134a}} = Q_{\text{evap1}} / (h[1] - h[4])$

Mass Flowrate of R23 $m_{\text{R23}} = Q_{\text{evap2}} / (h[5] - h[8])$